Lecture 4 (*February 24*)

Reading: Section 2.6
Viewing: Lecture 4

Let us return to Gaussian elimination, in the particular case of a square $n \times n$ matrix. For simplicity, let us assume that the algorithm doesn't require any rows to be exchanged (which is the case for most, but not all matrices). Then under this hypothesis, the algorithm goes as follows:

- first step: add a multiple λ_{21} of the first row to the second, then a multiple λ_{31} of the first row to the third, ... then a multiple λ_{n1} of the first row to the n–th row
- second step: add a multiple λ_{32} of the second row to the third, then a multiple λ_{42} of the second row to the fourth row, ... then a multiple λ_{n2} of the second row to the n–th row
- ...
- $(n-2)$–th step: add a multiple $\lambda_{n-1,n-2}$ of the $(n-2)$–th row to the $(n-1)$–th row, then add a multiple $\lambda_{n,n-2}$ of the $(n-2)$–th row to the n–th row
- $(n-1)$–th step: add a multiple $\lambda_{n,n-1}$ of the $(n-1)$–th row to the n–th row.

As we have seen in Lecture 2, each step in the algorithm above involves multiplying A on the left by an elimination matrix. As a formula, this states that:

$$\underbrace{E_{n,n-1}^{(\lambda_{n,n-1})}}_{(n-1)\text{–th step}} \underbrace{E_{n,n-2}^{(\lambda_{n,n-2})} E_{n-1,n-2}^{(\lambda_{n-1,n-2})}}_{(n-2)\text{–th step}} \underbrace{E_{n,n-3}^{(\lambda_{n,n-3})} E_{n-1,n-3}^{(\lambda_{n-1,n-3})} E_{n-2,n-3}^{(\lambda_{n-2,n-3})}}_{(n-3)\text{–th step}} \cdots$$
$$\cdots \underbrace{E_{n2}^{(\lambda_{n2})} \cdots E_{42}^{(\lambda_{42})} E_{32}^{(\lambda_{32})}}_{\text{second step}} \underbrace{E_{n1}^{(\lambda_{n1})} \cdots E_{31}^{(\lambda_{31})} E_{21}^{(\lambda_{21})}}_{\text{first step}} A = U \quad (30)$$

where U is a matrix in row echelon form. Because we are in the square matrix case, U is also a square matrix, so row echelon form is the same thing as **upper triangular**:

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} & u_{15} \\ 0 & u_{22} & u_{23} & u_{24} & u_{25} \\ 0 & 0 & u_{33} & u_{34} & u_{35} \\ 0 & 0 & 0 & u_{44} & u_{45} \\ 0 & 0 & 0 & 0 & u_{55} \end{bmatrix}$$

(the matrix above is how a 5×5 upper triangular matrix looks like). The opposite notion, where a square matrix has all its non-zero entries on or underneath the diagonal, is called **lower triangular**.

Fact 4. *Each elimination matrix $E_{ij}^{(\lambda)}$ is a lower triangular matrix, and its inverse is given by:*

$$\left(E_{ij}^{(\lambda)}\right)^{-1} = E_{ij}^{(-\lambda)} \quad (31)$$

Indeed, let's establish formula (31) in the case of 2×2 matrices (so $i = 2$, $j = 1$):

$$\begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -\lambda & 1 \end{bmatrix}$$

which is an easy computation to prove by hand. The proof of (31) for general $n \times n$ matrices is similar, and you can think about it, if you'd like.

Using formula (31), let's convert (30) into a formula for A. The first step to doing so is to multiply equation (30) on the left by the inverse of $E_{n,n-1}^{(\lambda_{n,n-1})}$, which according to (31) is precisely $E_{n,n-1}^{(-\lambda_{n,n-1})}$:

$$\left(E_{n,n-2}^{(\lambda_{n,n-2})} E_{n-1,n-2}^{(\lambda_{n-1,n-2})}\right)\left(E_{n,n-3}^{(\lambda_{n,n-3})} E_{n-1,n-3}^{(\lambda_{n-1,n-3})} E_{n-2,n-3}^{(\lambda_{n-2,n-3})}\right) \dots \left(E_{n1}^{(\lambda_{n1})} \dots E_{31}^{(\lambda_{31})} E_{21}^{(\lambda_{21})}\right) A =$$
$$= \left(E_{n,n-1}^{(\lambda_{n,n-1})}\right)^{-1} U = E_{n,n-1}^{(-\lambda_{n,n-1})} U$$

Then multiply the equation above on the left by the inverse of $E_{n,n-2}^{(\lambda_{n,n-2})} E_{n-1,n-2}^{(\lambda_{n-1,n-2})}$, which according to (27) and (31) is precisely $E_{n-1,n-2}^{(-\lambda_{n-1,n-2})} E_{n,n-2}^{(-\lambda_{n,n-2})}$. We obtain:

$$\left(E_{n,n-3}^{(\lambda_{n,n-3})} E_{n-1,n-3}^{(\lambda_{n-1,n-3})} E_{n-2,n-3}^{(\lambda_{n-2,n-3})}\right) \dots \left(E_{n1}^{(\lambda_{n1})} \dots E_{31}^{(\lambda_{31})} E_{21}^{(\lambda_{21})}\right) A =$$
$$= \left(E_{n,n-2}^{(\lambda_{n,n-2})} E_{n-1,n-2}^{(\lambda_{n-1,n-2})}\right)^{-1} \left(E_{n,n-1}^{(-\lambda_{n,n-1})}\right) U = \left(E_{n-1,n-2}^{(-\lambda_{n-1,n-2})} E_{n,n-2}^{(-\lambda_{n,n-2})}\right)\left(E_{n,n-1}^{(-\lambda_{n,n-1})}\right) U$$

We repeat this multiplication procedure until we get rid of all the matrices to the left of A, yielding:

$$A = \left(E_{21}^{(-\lambda_{21})} E_{31}^{(-\lambda_{31})} \dots E_{n1}^{(-\lambda_{n1})}\right) \dots \left(E_{n-1,n-2}^{(-\lambda_{n-1,n-2})} E_{n,n-2}^{(-\lambda_{n,n-2})}\right)\left(E_{n,n-1}^{(-\lambda_{n,n-1})}\right) U \tag{32}$$

Now here's the kicker: each $E_{ij}^{(\lambda)}$ is a lower triangular matrix, with all 1's on the diagonal. It is easy to see that a product of such lower triangular matrices is still lower triangular, for example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -5 & 3 & 1 \end{bmatrix}$$

Therefore, the product of elimination matrices in (32), namely:

$$L = \left(E_{21}^{(-\lambda_{21})} E_{31}^{(-\lambda_{31})} \dots E_{n1}^{(-\lambda_{n1})}\right) \dots \left(E_{n-1,n-2}^{(-\lambda_{n-1,n-2})} E_{n,n-2}^{(-\lambda_{n,n-2})}\right)\left(E_{n,n-1}^{(-\lambda_{n,n-1})}\right) \tag{33}$$

is still lower triangular, and moreover its diagonal entries are all 1. We conclude that:

$$\boxed{A = LU} \tag{34}$$

where L is a lower triangular matrix and U is an upper triangular matrix. Therefore, we summarize this whole discussion by saying that: **Gaussian elimination on a square matrix is equivalent to writing it as a product of a lower and an upper triangular matrix** (assuming there are no row exchanges). Moreover, if A is non-singular, then the matrices L and U in (34) are unique.

Let's do an example, namely Gaussian elimination of the matrix:

$$A = \begin{bmatrix} 2 & 4 & 1 \\ -4 & -5 & 0 \\ -2 & 5 & 6 \end{bmatrix}$$

First, let's add 2 times the first row to the second row, and we get:

$$E_{21}^{(2)}A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ -4 & -5 & 0 \\ -2 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 2 \\ -2 & 5 & 6 \end{bmatrix}$$

Then, let us add 1 times the first row to the third row, and we get:

$$E_{31}^{(1)}E_{21}^{(2)}A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 2 \\ -2 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 2 \\ 0 & 9 & 7 \end{bmatrix}$$

Finally, we add -3 times the second row to the third row, and we get:

$$E_{32}^{(-3)}E_{31}^{(1)}E_{21}^{(2)}A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 2 \\ 0 & 9 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix} =: U \tag{35}$$

Since the matrix on the right is upper triangular, hence in row echelon form, we are done with Gaussian elimination! To write A in the form (34), we need to isolate A in the left-hand side of (35). To do so, we must successively multiply the equation above with the inverses of the elimination matrices $E_{32}^{(-3)}$, $E_{31}^{(1)}$ and $E_{21}^{(2)}$, in this order. To this end, we have:

$$E_{31}^{(1)}E_{21}^{(2)}A = E_{32}^{(3)}U$$

$$E_{21}^{(2)}A = E_{31}^{(-1)}E_{32}^{(3)}U$$

$$A = E_{21}^{(-2)}E_{31}^{(-1)}E_{32}^{(3)}U$$

If we set:

$$L = E_{21}^{(-2)}E_{31}^{(-1)}E_{32}^{(3)} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix} \tag{36}$$

then we conclude that $A = LU$ holds with U and L given by (35) and (36), respectively.

A **diagonal** matrix is a square matrix, all of whose non-zero entries are on the diagonal:

$$D = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \tag{37}$$

A diagonal matrix is invertible if and only if all the diagonal entries d_i's are non-zero, and in this case its inverse is given by:

$$D^{-1} = \begin{bmatrix} \frac{1}{d_1} & 0 & \dots & 0 \\ 0 & \frac{1}{d_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{d_n} \end{bmatrix} \tag{38}$$

VERY IMPORTANT: If a matrix is not diagonal, then its inverse is **NOT** obtained by simply inverting all of its entries; for example you could think about the matrix all of whose entries are 1.

In the LU factorization (34), we note that L has 1's on the diagonal, while U is not required to. This is simply a matter of convention because if you'd like, you can write U as a diagonal matrix times an upper triangular matrix with 1's on the diagonal, for example:

$$\begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & \frac{1}{2} \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix} \tag{39}$$

Basically, this is done by writing $U = D\tilde{U}$, where D is the diagonal matrix with the same diagonal entries as the upper triangular matrix U. Then $\tilde{U}$ is forced to equal $D^{-1}U$, with the inverse D^{-1} given by formula (38). In these circumstances, $\tilde{U}$ will be upper triangular with 1's on the diagonal.

Definition 2. *The LU factorization of a matrix A consists of the unique matrices U and L (the former upper and the latter lower triangular with 1's on the diagonal) such that $A = LU$ holds.*

Similarly, the LDU factorization of a matrix A consists of the unique matrices U, D and L (where D is diagonal, while U and L are upper/lower triangular with 1's on the diagonal) such that $A = LDU$.

Note that if you have the LU factorization, you can easily get the LDU factorization, essentially by applying procedure (39). Let's end with a remark: suppose you do Gaussian elimination on a square matrix A, and you bring it in the form (32). Now you may be taken aback by the task of actually having to perform the multiplication (33) in order to compute L. However, a great fact about lower triangular matrices is the following formula:

$$\left(E_{21}^{(-\lambda_{21})} \dots E_{n1}^{(-\lambda_{n1})}\right) \dots \left(E_{n-1,n-2}^{(-\lambda_{n-1,n-2})} E_{n,n-2}^{(-\lambda_{n,n-2})}\right)\left(E_{n,n-1}^{(-\lambda_{n,n-1})}\right) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\lambda_{21} & 1 & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ -\lambda_{n-1,1} & -\lambda_{n-1,2} & \dots & 1 & 0 \\ -\lambda_{n1} & -\lambda_{n2} & \dots & -\lambda_{n,n-1} & 1 \end{bmatrix}$$

Note that this formula <u>only</u> holds because of the specific order in which the matrices $E_{ij}^{(-\lambda)}$ were multiplied in the left-hand side. If you had multiplied the matrices in (almost) any other order, you would not have obtained a result as nice as the right-hand side.